# David Pelosi

**PhD Candidate - Astroparticle and Space Physics**

## CONTACT INFORMATION


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## CURRENT POSITION

**PhD Candidate** at **Department of Physics and Geology, Università degli Studi di Perugia — UniPG**, Perugia, Italy.

Thesis title: *The PgLis Model: A forecasting model for Galactic Cosmic Rays Variability and Orbital Radiation Risk Assessment in Space Missions*

Supervisor: Nicola Tomassetti

Co-Supervisor: Miguel Orcinha

End of PhD grant: 31 October 2026.

## EDUCATION

**Department of Physics and Geology, Università degli Studi di Perugia — UniPG, Perugia, Italy**

**M.Sc.** in **Physics** **27 Apr 2023**

Specialization: Astrophysics and Astroparticle Physics

Thesis title:*A Numerical Model for the Transport of Cosmic Rays in the Heliosphere.*

Supervisor: Nicola Tomassetti

Grade: 110/110 *cum laude*

**B.Sc.** in **Physics** **11 Jun 2020**

Thesis title:*Development of a web application for monitoring cosmic radiation and solar activity.*

Supervisor: Nicola Tomassetti

Co-Supervisor: Matteo Duranti

Grade: 110/110

**Liceo Scientifico Galeazzo Alessi, Perugia, Italy**

**High School Diploma** **Jul 2015**

Grade: 100/100

## TEACHING EXPERIENCE

**Università degli Studi di Perugia — UniPG**, Perugia, Italy

**Physics (Mechanics and Electromagnetism)**

Classes for *B.Sc. in Civil and Environmental Engineering*:

**Oct 2022–May 2023** as a Teaching Collaborator
**Oct 2021– Jun 2022** as a Teaching Collaborator
**Mar 2021– May 2021** as a Teaching Collaborator

Classes for *M.Sc. in Building Engineering and Architecture*:

**Mar 2022–Jun 2022** as a Teaching Collaborator
**Mar 2021 -Jun 2021** as a Teaching Collaborator

## FELLOWSHIPS, GRANTS & AWARDS

**Invited Visiting Researcher** **Jan 2026 — March 2026**

Location: Laboratório de Instrumentação e Física Experimental de Partículas (LIP), Lisbon, Portugal

Contact person: Fernando José de Carvalho Barão

**Scientific Research Grant** **Mar 2025 — Jun 2025**

Title: SCOSTEP Visiting Scholar (SVS) program 2025

Position: Visiting Researcher

Location: Space Physics and Astronomy Research Unit at the University of Oulu, Finland

Funding: Scientific Committee on Solar-Terrestrial Physics (SCOSTEP).

**Scientific Research Grant** **Mar 2025 — Jun 2025**

Title: Advancing Galactic Cosmic Ray modeling and its implications for atmospheric chemistry and climate dynamics
Position: Visiting Researcher
Location: Space Physics and Astronomy Research Unit at the University of Oulu, Finland
Funding: Geomagnetic Excursions/Reversals: Assessment of Climate Impacts in Silico (GERACLIS) funded by the Research Council of Finland for 2023-2027 and hosted by the University of Oulu in consortium with Finnish Meteorological Institute.
Contact persons: Ilya Usoskin, Pauli Väisänen

**The 2025 Outstanding Paper Award for Young Scientists** **Jun 2026**

for the paper entitled "A forecasting framework for galactic cosmic ray flux in space weather applications"in *Advances in Space Research, Volume 76, Issue 9, 1 November 2025, Pages 5700-5713*.

**Best presentation award IFAE 2026** **Apr 2026**

Best oral presentation award of the session "Astroparticle and Cosmology" at the Incontri di Fisica delle Alte Energie — IFAE 2026, Bologna, Italy, 8-10 April 2026.

**Franco Mariani Award** **Nov 2024**

The "Premio Franco Mariani" is awarded by the Space Weather Italian Community (SWICO) for the best master's thesis in Space Weather research.

**Ph.D. Grant** **Nov 2023 — Oct 2026**

Topic: Phenomenological study of energetic charged radiation in space
Position: PhD student
Funding: Università degli Studi di Perugia — UniPG

**Postgraduate Research Fellow** **Jan 2022 — Oct 2023**

Topic: Implementation of an efficient model for predicting the flux of galactic cosmic rays.
Position: M.Sc. student
Project: Comprehensive spAce wEather Studies for the ASPIS prototype Realization (CAESAR) project, supported by the Italian Space Agency (ASI) and the National Institute for Astrophysics (INAF).

## Refereed Journal Publications

The following papers were published in the following international peer-reviewed journals: **Physical Review Letters Advances in Space Research** and **Il Nuovo Cimento C**. A complete reference list of the publications can be found on the INSPIRE database **here**.

13. D. Pelosi et al., "New model for long-term forecasting of galactic cosmic rays", Phys. Rev. D 114, 063005 (2026),
doi: 10.1103/npfk-lwy9

12. P. Väisänen et al., "Near-Earth Cosmic Ray Modulation Index $\phi$ and Neutron Monitor Cutoff Rigidities at Hourly Cadence: Toward an Improved Monitoring of Solar-Terrestrial Radiation Environment, Space Weather, and Space Climate", Journal of Geophysical Research: Space Physics, 131, 8 (2026),
doi: 10.1029/2026JA035445

11. A. Aceituno et al. (AMS Collaboration), "Properties of Heavy Cosmic Nuclei Phosphorus, Chlorine, Argon, Potassium, and Calcium: Results from the Alpha Magnetic Spectrometer", Phys. Rev. Lett. 136 (2026) 241002,
doi: 10.1016/j.asr.2025.08.022

10. D. Pelosi et al., "A forecasting framework for galactic cosmic ray flux in space weather applications", Advances in Space Research 76 (2025) 5700-5713,
doi: 10.1016/j.asr.2025.08.022

9. M. Aguilar et al. (AMS Collaboration), "Properties of Cosmic Lithium Isotopes Measured by the Alpha Magnetic Spectrometer", Phys. Rev. Lett. 134 (2025) 20, 201001, doi: 10.1103/PhysRevLett.134.201001

8. M. Aguilar et al. (AMS Collaboration), "Antiprotons and Elementary Particles over a Solar Cycle: Results from the Alpha Magnetic Spectrometer", Phys. Rev. Lett. 134 (2025) 051002, doi: 10.1103/PhysRevLett.134.051002

7. M. Aguilar et al. (AMS Collaboration), "Solar Modulation of Cosmic Nuclei over a Solar Cycle: Results from the Alpha Magnetic Spectrometer", Phys. Rev. Lett. 134 (2025) 051001, doi: 10.1103/PhysRevLett.134.051001

6. D. Pelosi et al. "Development of a web application for monitoring solar activity and cosmic radiation". Nuovo Cim.C 44 (2021), 1259. doi:10.1393/ncc/i2021-21097-2

## CONFERENCE PROCEEDINGS

5. D. Pelosi et al. "Modeling the heliospheric modulation of cosmic-ray particles and antiparticles in light of new multichannel data from AMS-02 in space". Proceedings of 39th International Cosmic Ray Conference (2025) — PoS(ICRC2025), 1371. doi:10.22323/1.501.1371

4. D. Pelosi et al. "Analysis of Galactic Cosmic Ray Variability and Time-Lagged Relation to Solar Activity". Proceedings of 39th International Cosmic Ray Conference (2025) — PoS(ICRC2025), 1353. doi:10.22323/1.501.1353

3. M. Orcinha, F. Barão, B. Bertucci, E. Fiandrini, D. Pelosi and N. Tomassetti, "Understanding variations of galactic energetic particles in the heliosphere: modelling and radiation hazard assessment", Cent. Eur. Astrophys. Bull. 48 (2024), (awaiting publication), The 28th European Cosmic Ray Symposium — ECRS 2024, Hvar, Croatia, 23-27 September 2024, doi: 10.48550/arXiv.2509.11837

2. D. Pelosi, F. Barão, B. Bertucci, E. Fiandrini, M. Orcinha, A. Reina Conde and N. Tomassetti, "Cross-correlation analysis for cosmic ray flux forecasting", EPJ Web Conf. 319 (2025) 13004, Roma International Conference on AstroParticle Physics — RICAP-24, Frascati, Italy, 23-27 September 2024, doi: 10.1051/epjconf/202531913004

1. D. Pelosi et al. "A web application for monitoring cosmic rays and solar activity". Proceedings of 37th International Cosmic Ray Conference (2021) — PoS(ICRC2021), 1259. doi:10.22323/1.395.1259

## CONFERENCE & WORKSHOP PRESENTATIONS

The speaker of each presentation is marked with an asterisk (*).

**Contributed talks at international conferences and workshops**

**7 Aug 2026** "The PGLis Model: A Tool for Long-Term Galactic Cosmic Ray Forecasting", F. Barão, B. Bertucci, E. Fiandrini, M. Orcinha, **D. Pelosi**, N. Tomassetti*, 1-9 August 2026, The 46th COSPAR Scientific Assembly, Florence, Italy

**7 Aug 2026** "Heliospheric and Earth's magnetosphere impact on cosmic radiation hazards in near Earth orbits", M. Orcinha*, F. Barão, B. Bertucci, E. Fiandrini, **D. Pelosi**, N. Tomassetti, 1-9 August 2026, The 46th COSPAR Scientific Assembly, Florence, Italy

**11 Jun 2026** "The near-Earth modulation (NEM) index in high cadence for study of global cosmic ray modulation and space climate", P. Väisänen*, S. Koldobskiy, N. Larsen, M. Similä, **D. Pelosi**, M. Orcinha, 9-12 June 2026, Space Climate 10 — Åland/Ahvenanmaa, Finland

**10 Apr 2026** "Il Modello PgLis: per la previsione a lungo termine dei raggi cosmici", **D. Pelosi***, F. Barão, B. Bertucci, E. Fiandrini, M. Orcinha, N. Tomassetti, 8-10 April 2026, Incontri di Fisica delle Alte Energie — IFAE 2026, Bologna, Italy

**30 Oct 2025** "Cosmic radiation near Earth: Impact of solar activity and the geomagnetic field in long-term modelling of dose from cosmic radiation in space", M. Orcinha*, F. Barão, B. Bertucci, E. Fiandrini, **D. Pelosi**, N. Tomassetti, 27-31 October 2025, European Space Weather Week — ESWW2025, Umeå, Sweden

**28 Oct 2025** "Solar Activity and Galactic Cosmic Rays: A Comprehensive Model for Long-Term Forecasting", **D. Pelosi***, F. Barão, B. Bertucci, E. Fiandrini, M. Orcinha, N. Tomassetti, 27-31 October 2025, European Space Weather Week — ESWW2025, Umeå, Sweden

**17 Jul 2025** "Modeling the heliospheric modulation of cosmic-ray in light of new data from AMS-02 in space", N. Tomassetti* and A. Reina Conde and B. Bertucci and **D. Pelosi** and E. Fiandrini, F. Barao and F. Faldi and M. Orcinha, 14-24 July 2025, International Cosmic-Ray Conference — ICRC2025, Geneva, Switzerland

**16 Jul 2025** "Temporal Evolution of the Daily Proton and Helium Fluxes with AMS-02", F. Faldi* and M. Orcinha and **D. Pelosi**, on Behalf of AMS-02 Collaboration, 14-24 July 2025, International Cosmic-Ray Conference — ICRC2025, Geneva, Switzerland

**28 Mar 2025** "Solar Activity and Galactic Cosmic Rays: A Comprehensive model for Long-term forecasting and Radiation Safety", **D. Pelosi***, F. Barão, B. Bertucci, E. Fiandrini, M. Orcinha, N. Tomassetti, 26-28 March 2025, Physics Days 2025, Oulu, Finland.

**27 Nov 2024** "An Effective and Predictive Model for the Long-Term Variations of Cosmic Rays in the Heliosphere", **D. Pelosi***, M. Orcinha, F. Barão, B. Bertucci, E. Fiandrini and N. Tomassetti, Space Weather Italian Community (SWICO) Congress — Italian Space Agency (ASI), Rome, Italy.

**8 Nov 2024** "Long-term variations of cosmic rays in the heliosphere: modelling, forecasting and radiation hazard assessment", M. Orcinha*, F. Barão, B. Bertucci, F. Faldi, E. Fiandrini, **D. Pelosi**, N. Tomassetti, P. Väisänen, European Space Weather Week 2024 — ESWW2024, Coimbra, Portugal

**23 Sep 2024** "An effective and predictive model for the long-term variations of Cosmic Rays in the Heliosphere", M. Orcinha*, F. Barão, B. Bertucci, E. Fiandrini, **D. Pelosi**, A. Reina Conde, N. Tomassetti, 28th European Cosmic Ray Symposium — ECRS2024, Hvar, Croatia

**14 Sep 2023** "An effective and predictive model for the long-term variations of Cosmic Rays in the Heliosphere", **D. Pelosi***, F. Barão, B. Bertucci, E. Fiandrini, M. Orcinha, A. Reina Conde, N. Tomassetti, 14 September 2023, 109 Congresso Nazionale, Società Italiana di Fisica, Salerno, Italy

**17 May 2023** "An Effective and Predictive Model for the Long-term Variations of Cosmic Rays in the Heliosphere", **D. Pelosi***, B. Bertucci, E. Fiandrini and N. Tomassetti, 15 - 17 May 2023, Space Environments Monitoring Workshop – SPACEMON 2023, ESA/ESTEC, Noordwijk, The Netherlands.

**Poster sessions**

**9 Jun 2026** "On the time lag between GCRs and solar activity: A Wavelet-Hilbert Analysis", **D. Pelosi*** and F. Barão and B. Bertucci and E. Fiandrini and M. Orcinha and N. Tomassetti, 9-12 June 2026, Space Climate 10 — Åland/Ahvenanmaa, Finland

**14 Jul 2025** "Analysis of Time Variability of Galactic Cosmic Rays", **D. Pelosi*** and F. Barão and B. Bertucci and F. Faldi and E. Fiandrini and M. Orcinha and N. Tomassetti and P. Väisänen and R. C. Venterea, 14-24 July 2025, International Cosmic-Ray Conference — ICRC2025, Geneva, Switzerland

**23 Sep 2024** "An effective and predictive model for the long-term variations of Cosmic Rays in the Heliosphere", **D. Pelosi***, F. Barão, B. Bertucci, E. Fiandrini, M. Orcinha, A. Reina Conde, N. Tomassetti, 23-27 September 2024, RICAP24, Frascati, Italy

**14 Apr 2024** "An effective and predictive model for the long-term variations of Cosmic Rays in the Heliosphere", **D. Pelosi***, F. Barão, B. Bertucci, E. Fiandrini, M. Orcinha, A. Reina Conde, N. Tomassetti, 14–19 April 2024, European Geosciences Union — EGU, Vienna, Austria

**12 Jul 2021** "A Web Application for Monitoring Cosmic Rays and Solar Activity", **D. Pelosi***, M. Duranti and N. Tomassetti, 12-23 July 2021, International Cosmic-Ray Conference — ICRC2021, Berlin, Germany

## ADVANCED EDUCATION

**Participation in international schools**

**27 Jul - 31 Jul 2026**
XIII International GEANT4 School.

Location: ELI Beamlines Facility, Czech Republic
Organizers: Laboratori Nazionali del Sud of INFN, ELI-ERIC in collaboration with COST Action CA21128 PROBONO.
Completed with evaluation passed

**9 Apr - 24 Apr 2026**
2nd CNCG School on Global Navigation Satellite Systems and related technologies.
Location: Telespazio, Rome, Italy
Organizers: Telespazio, CNIT - Consorzio Nazionale Interuniversitario per le Telecomunicazioni, Agenzia Spaziale Italiana.
Online sessions (9-21 Apr) and on-site laboratory activities (23-24 Apr).
Completed with evaluation passed

**29 Sept - 9 Oct 2025**
XVI INFN International School on Efficient Scientific Computing (ESC25) about "Architectures, tools and methodologies for the development of large-scale scientific computing applications".
Location: Bertinoro, Italy
Organizers: INFN, CNAF
Completed with evaluation passed

**25 Aug - 10 Sept 2024**
The ESA-FAIR Space Radiation School 2024.
Location: Darmstadt, Germany
Organizers: European Space Agency and GSI/FAIR

**13 - 17 May 2024**
INFN School of Statistics
Organizers: INFN and University "Federico II", Naples. Location: Paestum, Italy

**29 Apr - 3 May 2024**
ESPD summerschool.
Theme: Energisation and heating in the solar plasma.
Organizers: European Solar Physics Division (ESPD) of the European Physical Society (EPS).
Location: Dubrovnik, Croatia

**23 - 27 Oct 2023**
International School on Open Science Cloud (SOSC 2023).
Theme: Computing Models for Scientific Experiments.
Location: Perugia, Italy
Organizers: INFN, University of Bologna and the University of Perugia.
Completed with evaluation passed

**16 - 18 Sep 2021**
NiPS-EnABLES Summer School 2021
Theme: Powering the Internet of Things.
Location: Perugia, Italy
Organizers: NiPS Laboratory

## KNOWLEDGE TRANSFER

**Invited talks at academic institutions**

**12 March 2026** "The PGLis model and radiation dose studies", LIP — Laboratório de Instrumentação e Física Experimental de Partículas, Lisbon, Portugal

**8 Apr 2025** "Long-term forecasting of Galactic Cosmic Rays and radiation dose studies", Space Physics Seminars, Space Physics and Astronomy Research Unit at the University of Oulu, Finland

**28 Nov 2024** "Plans for Future Research on GCR and Radiation Dose Studies," talk given at the award ceremony of the "Premio Franco Mariani," Space Weather Italian Community (SWICO) Congress — Italian Space Agency (ASI), Rome, Italy.

**Outreach initiatives**

**Jun 2019 – Jun 2021**
President, Italian Association of Physics Students (AISF) Local Committee of Perugia.

**26 Sep 2025** Participation in "Particelle nella Nebbia", made for SHARPER — the EU's Researcher's Night 2025, organized by INFN and Dept. of Physics & Geology of the University of Perugia, Italy

**26 Nov 2024** Participation in "International Cosmic Day" (ICD-2024), organized by INFN-Perugia and Dept. of Physics & Geology of the University of Perugia, Italy. Nationally coordinated by the network OCRA — Outreach Cosmic Ray Activities

**30 Sep 2023** Participation in "Particelle nella Nebbia", made for SHARPER — the EU's Researcher's Night 2023, organized by INFN and Dept. of Physics & Geology of Department of Physics and Geology, Università degli Studi di Perugia — UniPG, Perugia, Italy

## Scientific projects

**AMS — Alpha Magnetic Spectrometer** **Nov 2023 - present**
Role: PhD student

**CAESAR** **Jan 2022 - Oct 2023**
Title: CAESAR — Comprehensive spAce wEather Studies for the ASPIS prototype Realization
Funding: INAF-ASI — Instituto Nazionale di Astrofisica, Agenzia Spaziale Italiana, Italy, ref. 2020-35-HH.0
Role: M.Sc. student

## Other activities

**Current Scientific positions & Memberships**

**2023 - present** — **Associated Member** of the Istituto Nazionale di Fisica Nucleare — INFN, Perugia, Italy

**2023 - present** — **Member** of AMS — international scientific collaboration Alpha Magnetic Spectrometer

**2023 - present** — **Member** of CERN — the European Organization for Nuclear Research

**2026 - present** — **Consultant** for BEAMIDE — radiation simulation and detection software company

**Service**

**2026** — Reviewer for the journal **Advances in Space Research** (ASR).

**2023 - present** — AMS Collaboration — shifter for the TEE sub-detector group (TRD, ACC, TRK).

**2024 – present** — Responsible for coordinating the Italian side of the AMS Collaboration in the organization and management of shifts for the TEE sub-detector group (TRD, ACC, TRK).

**Academic role**

**Jan 2026 - present** — Representative of the PhD students of the Department of Physics and Geology, Università degli Studi di Perugia — UniPG, Perugia, Italy

**Apr 2026 – present** — "Cultore della materia" (Teaching Collaborator) for the course of Physics in the Earth Sciences and Environmental Sciences programs at Department of Physics and Geology, Università degli Studi di Perugia — UniPG, Perugia, Italy

## Hardware & Software Skills

**Programming & Scripting Languages**
> C, C++
> Python
> UNIX shell scripting (bash, zsh)
> Markdown (GitHub, Emacs)

**Parallelization and Communication Technologies**
> OpenMP
> TBB
> Threads

**Signal Processing**
> Empirical Mode Decomposition
> Wavelet Transform
> Fourier Transform
> Hilbert Transform

**Data Analysis**
> Wolfram Mathematica
> ROOT — Data Analysis Framework, CERN
> TMVA — Toolkit for Multivariate Data Analysis with ROOT
> Numpy
> Pandas

**Data Structures**

> ROOT files (C++ Object Oriented Storage)
> JSON
> CSV (and similar)

**Package Management Software**

> Homebrew
> Apt (apt-get) — Advanced Packaging Tool

**Version Control Software**

> DVCS (Git) — GitHub, GitLab

**Text Editing and Productivity Software**

> Vim, Emacs
> Visual Studio Code
> TeX
> Microsoft Office, OpenOffice.org, LibreOffice, Google Docs

**Operating Systems**

> Apple OSX
> Linux

## Language Skills

Italian – native
English – fluent